OpsPilot — Complete Solution

> The whole project, every file — nothing to copy from anywhere else. Use it to **check your work**, not to skip the thinking. Build from the story first; read the hints if you get stuck. All of this code runs against the example services with no external setup.

The final layout

```
<repo folder>/
├── servers/
│   ├── runbook_server.py
│   └── status_server.py
├── src/
│   ├── models/
│   │   ├── __init__.py
│   │   └── llm.py
│   ├── utils/
│   │   ├── __init__.py
│   │   └── logger.py
│   ├── __init__.py
│   ├── agent.py
│   ├── config.py
│   ├── exceptions.py
│   ├── server_pool.py
│   ├── servers_config.py
│   └── tool_bridge.py
├── tests/
│   ├── integration/
│   │   └── test_agent.py
│   └── unit/
│       ├── test_config.py
│       └── test_tool_bridge.py
├── .env.example
├── .gitignore
├── LICENSE
├── main.py
└── requirements.txt
```

The project files you make once

`requirements.txt`

```
mcp
openai
python-dotenv
pydantic
```

`.env.example` (copy to `.env`, put your real key in `.env`)

```
OPENAI_API_KEY=sk-your-key-here
LOG_LEVEL=INFO
MCP_CONNECT_TIMEOUT=10
MCP_MAX_ITERATIONS=6
```

`.gitignore`

```
.env
.venv/
__pycache__/
*.pyc
```

Every folder under `src/` gets an **empty** `__init__.py`: `src/__init__.py`, `src/models/__init__.py`, `src/utils/__init__.py`. They hold nothing — the file just needs to exist so Python treats the folder as a package.

`LICENSE` — MIT (fill in the year and your name):

```
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```

```
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SOFTWARE.
```

The two example services

```
# servers/status_server.py
# A minimal MCP service for service health. The agent launches this as a
# subprocess and discovers its tools — it never imports this file.
import sys
from mcp.server.mcpserver import MCPServer

mcp = MCPServer("status-server")

STATUS = {
    "checkout": "DOWN since 14:02 UTC - 503 rate 88%.",
    "search": "healthy - p95 210ms.",
}
DEPLOYS = {
    "checkout": "v812 deployed 13:58 UTC by sara (14 commits).",
    "search": "v77 deployed yesterday.",
}

@mcp.tool()
def get_service_status(service: str) -> str:
    """Get the current health status of one service by name."""
    if service not in STATUS:
        return f"Error: no service named {service}"
    return STATUS[service]

@mcp.tool()
def recent_deploys(service: str) -> str:
    """Get the most recent deploy for one service by name."""
    if service not in DEPLOYS:
        return f"Error: no service named {service}"
    return DEPLOYS[service]

if __name__ == "__main__":
    print("status server starting", file=sys.stderr)
    mcp.run(transport="stdio")
```

```
# servers/runbook_server.py
# A minimal MCP service for runbooks. Independent of the status service —
# its own domain, its own tools.
import sys
from mcp.server.mcpserver import MCPServer

mcp = MCPServer("runbook-server")

RUNBOOKS = {
    "rb-checkout": "Checkout 503s: roll back the deploy, then page payments.",
    "rb-search": "Search latency: clear the query cache and add replicas.",
}

@mcp.tool()
def search_runbooks(query: str) -> str:
    """Search runbooks by keyword; return matching runbook ids."""
    hits = []
    for runbook_id, text in RUNBOOKS.items():
        if query.lower() in text.lower() or query.lower() in runbook_id:
            hits.append(f"{runbook_id}: {text[:30]}")
    if not hits:
        return "No matching runbooks."
    return "\n".join(hits)

@mcp.tool()
def read_runbook(runbook_id: str) -> str:
    """Read the full text of one runbook by its id."""
    if runbook_id not in RUNBOOKS:
        return f"Error: no runbook with id {runbook_id}"
    return RUNBOOKS[runbook_id]

if __name__ == "__main__":
    print("runbook server starting", file=sys.stderr)
    mcp.run(transport="stdio")
```

Settings, errors, logging

```
# src/exceptions.py
class MissingConfigError(Exception):
    """Raised when a required setting is missing from the environment."""
    pass

class ServerUnreachableError(Exception):
    """Raised when an MCP service cannot be connected to in time."""
```

```
    pass

class MaxIterationsExceeded(Exception):
    """Raised when the ReAct loop runs more steps than its hard limit."""
    pass
```

```
# src/config.py
# Loads settings from .env into a checked Config object (the Doc01 pattern).
import os
from dataclasses import dataclass
from dotenv import load_dotenv
from src.exceptions import MissingConfigError

@dataclass
class Config:
    openai_api_key: str
    log_level: str
    connect_timeout: float
    max_iterations: int

def require_env(key: str) -> str:
    value = os.getenv(key)
    if value is None or value == "":
        raise MissingConfigError(
            f"Required environment variable is missing: {key}"
        )
    return value

def load_config() -> Config:
    load_dotenv()
    api_key = require_env("OPENAI_API_KEY")
    log_level = os.getenv("LOG_LEVEL", "INFO")
    connect_timeout = float(os.getenv("MCP_CONNECT_TIMEOUT", "10"))
    max_iterations = int(os.getenv("MCP_MAX_ITERATIONS", "6"))
    return Config(
        openai_api_key=api_key,
        log_level=log_level,
        connect_timeout=connect_timeout,
        max_iterations=max_iterations,
    )
```

```
# src/utils/logger.py
# One logging setup for every file. Logs to STDERR on purpose: an MCP
# client can carry protocol traffic on stdout.
import logging
```

```
import os
import sys

def get_logger(name: str) -> logging.Logger:
    logger = logging.getLogger(name)
    if not logger.handlers:
        level_name = os.getenv("LOG_LEVEL", "INFO")
        level = getattr(logging, level_name.upper(), logging.INFO)
        handler = logging.StreamHandler(sys.stderr)
        handler.setLevel(level)
        log_format = "%(asctime)s %(name)s %(levelname)s %(message)s"
        handler.setFormatter(logging.Formatter(log_format))
        logger.addHandler(handler)
        logger.setLevel(level)
    return logger
```

The schema bridge

```
# src/tool_bridge.py
# Adapters between MCP's shapes and OpenAI's, so the loop never has to know
# either library's exact types.
from src.utils.logger import get_logger

def one_openai_tool(namespaced_name, description, input_schema):
    return {
        "type": "function",
        "function": {
            "name": namespaced_name,
            "description": description,
            "parameters": input_schema,
        },
    }

def mcp_tools_to_openai_schema(named_tools):
    logger = get_logger(__name__)
    schemas = []
    for namespaced_name, tool in named_tools:
        description = tool.description
        if description is None:
            description = ""
        schemas.append(
            one_openai_tool(namespaced_name, description, tool.input_schema)
        )
    logger.info(f"converted {len(schemas)} MCP tools to OpenAI schema")
    return schemas
```

```
def extract_text(call_result):
    parts = []
    for block in call_result.content:
        text = getattr(block, "text", None)
        if text is not None:
            parts.append(text)
    return "\n".join(parts)
```

The server pool (connect, discover, route, survive)

```
# src/server_pool.py
# MCPServerPool: connect to every configured service, discover and namespace
# its tools, and route calls back to the right service. Resilient: a service
# that can't connect is skipped; a tool call that fails mid-task returns a
# clear error string instead of crashing the loop.
import asyncio
from contextlib import AsyncExitStack

from mcp import ClientSession, StdioServerParameters
from mcp.client.stdio import stdio_client

from src.tool_bridge import extract_text, mcp_tools_to_openai_schema
from src.utils.logger import get_logger

class MCPServerPool:
    def __init__(self, servers, connect_timeout=10.0):
        self.servers = servers
        self.connect_timeout = connect_timeout
        self.stack = AsyncExitStack()
        self.sessions = {}
        self.routing = {}
        self.named_tools = []
        self.logger = get_logger(__name__)

    async def _connect_one(self, server):
        # The context managers are entered in THIS (the main) task, never
        # inside asyncio.wait_for: wait_for runs its coroutine in a new
        # task, and a context entered in one task cannot be closed in
        # another. Only initialize() -- which can hang -- is timed out.
        params = StdioServerParameters(
            command=server["command"], args=server["args"]
        )
        read, write = await self.stack.enter_async_context(
            stdio_client(params)
```

```
        )
        session = await self.stack.enter_async_context(
            ClientSession(read, write)
        )
        await asyncio.wait_for(
            session.initialize(), timeout=self.connect_timeout
        )
        return session

    async def connect_all(self):
        for server in self.servers:
            name = server["name"]
            try:
                session = await self._connect_one(server)
            except Exception as e:
                self.logger.warning(
                    f"[{name}] unreachable: {e}, continuing without it"
                )
                continue
            self.sessions[name] = session
            listed = await session.list_tools()
            for tool in listed.tools:
                namespaced = name + "__" + tool.name
                self.routing[namespaced] = (name, tool.name)
                self.named_tools.append((namespaced, tool))
            self.logger.info(f"[{name}] connected, {len(listed.tools)} tools")

    def openai_tools(self):
        return mcp_tools_to_openai_schema(self.named_tools)

    async def call_tool(self, namespaced_name, arguments):
        if namespaced_name not in self.routing:
            return f"Error: no tool named {namespaced_name} is available"
        server_name, tool_name = self.routing[namespaced_name]
        session = self.sessions[server_name]
        try:
            result = await session.call_tool(tool_name, arguments)
        except Exception as e:
            self.logger.warning(f"[{server_name}] call failed: {e}")
            return f"Error: tool {namespaced_name} failed: {e}"
        return extract_text(result)

    async def aclose(self):
        await self.stack.aclose()
```

> **The one real async trap:** do not write `await asyncio.wait_for(self._connect_one(server), ...)`. `wait_for` runs its coroutine in a **new task**, so the `stdio_client` / `ClientSession` contexts get entered there; closing them later in the main task raises `RuntimeError: Attempted to exit cancel scope in a different task`. Enter the contexts in the main task, and time out only `initialize()`.

The OpenAI call and the ReAct loop

```
# src/models/llm.py
# The only file that talks to the OpenAI SDK.
import openai
from openai import OpenAI

from src.config import Config

MODEL = "gpt-4o-mini"

def create_client(config: Config) -> OpenAI:
    return OpenAI(api_key=config.openai_api_key)

def check_api_key_at_startup(client: OpenAI) -> bool:
    try:
        client.chat.completions.create(
            model=MODEL,
            messages=[{"role": "user", "content": "ping"}],
            max_tokens=1,
        )
    except openai.AuthenticationError:
        return False
    return True

def ask_model(client, messages, tools):
    response = client.chat.completions.create(
        model=MODEL, messages=messages, tools=tools
    )
    return response.choices[0].message
```

```
# src/agent.py
# The ReAct loop: think (ask the model) -> act (call a discovered tool via
# the pool) -> observe (feed the result back). Bounded by max_iterations.
import json
from dataclasses import dataclass
```

```
from src.exceptions import MaxIterationsExceeded
from src.models.llm import ask_model
from src.utils.logger import get_logger

@dataclass
class AgentResult:
    answer: str
    steps: list

def build_messages(task):
    system = (
        "You are an on-call assistant. Use the available tools to answer "
        "the question. When you have enough information, answer plainly."
    )
    return [
        {"role": "system", "content": system},
        {"role": "user", "content": task},
    ]

async def run_agent(task, pool, client, max_iterations=6):
    logger = get_logger(__name__)
    messages = build_messages(task)
    tools = pool.openai_tools()
    steps = []

    for step_number in range(max_iterations):
        reply = ask_model(client, messages, tools)
        if not reply.tool_calls:
            logger.info("agent finished with a final answer")
            return AgentResult(answer=reply.content, steps=steps)

        messages.append({
            "role": "assistant",
            "content": reply.content,
            "tool_calls": _tool_calls_as_dicts(reply.tool_calls),
        })
        for call in reply.tool_calls:
            name = call.function.name
            arguments = json.loads(call.function.arguments)
            logger.info(f"step {step_number}: calling {name} {arguments}")
            observation = await pool.call_tool(name, arguments)
            steps.append({"tool": name, "args": arguments,
                          "observation": observation})
            messages.append({
                "role": "tool",
                "tool_call_id": call.id,
```

```
                "content": observation,
            })

    raise MaxIterationsExceeded(
        f"agent did not finish within {max_iterations} steps"
    )

def _tool_calls_as_dicts(tool_calls):
    out = []
    for call in tool_calls:
        out.append({
            "id": call.id,
            "type": "function",
            "function": {
                "name": call.function.name,
                "arguments": call.function.arguments,
            },
        })
    return out
```

The config and the entry point

```
# src/servers_config.py
# The services to connect to, as plain data. OpsPilot never hardcodes these
# services' TOOLS — only how to launch them.
SERVERS = [
    {"name": "status", "command": "python",
     "args": ["servers/status_server.py"]},
    {"name": "runbooks", "command": "python",
     "args": ["servers/runbook_server.py"]},
]
```

```
# main.py
# The demo you run: connect to the configured services, discover their
# tools, answer one on-call question, and print each step.
import asyncio
import sys

from src.agent import run_agent
from src.config import load_config
from src.exceptions import MissingConfigError
from src.models.llm import create_client
from src.server_pool import MCPServerPool
from src.servers_config import SERVERS
```

```
TASK = ("Is checkout down right now, and what does the runbook say to do "
        "about it?")

async def run_demo():
    try:
        config = load_config()
    except MissingConfigError as e:
        print(f"Setup problem: {e}")
        sys.exit(1)

    client = create_client(config)
    pool = MCPServerPool(SERVERS, connect_timeout=config.connect_timeout)
    await pool.connect_all()
    print(f"Connected servers: {list(pool.sessions.keys())}")
    print(f"Tools discovered: {len(pool.named_tools)}")
    try:
        print(f"\nTask: {TASK}")
        result = await run_agent(
            TASK, pool, client, max_iterations=config.max_iterations
        )
        for step in result.steps:
            print(f"  -> {step['tool']} {step['args']}")
        print(f"Answer: {result.answer}")
    finally:
        await pool.aclose()

def main():
    asyncio.run(run_demo())

if __name__ == "__main__":
    main()
```

Expected output (`python main.py` , with a real key) — the incident question routes to both services:

```
Connected servers: ['status', 'runbooks']
Tools discovered: 4

Task: Is checkout down right now, and what does the runbook say to do about it?
  -> status__get_service_status {'service': 'checkout'}
  -> runbooks__search_runbooks {'query': 'checkout'}
  -> runbooks__read_runbook {'runbook_id': 'rb-checkout'}
Answer: Checkout has been down since 14:02 (503 rate 88%). The runbook says
to roll back the deploy, then page payments.
```

The tests

```
# tests/unit/test_config.py
# Checks the settings code with no network. Each case prints OK or WRONG.
# It changes only a made-up setting, so your real .env is never touched.
# Run from the repo root: python -m tests.unit.test_config
import os

from src.config import load_config, require_env
from src.exceptions import MissingConfigError

TEST_SETTING = "OPSPILOT_TEST_SETTING"

def test_missing_setting() -> None:
    os.environ.pop(TEST_SETTING, None)
    result = "WRONG"
    try:
        require_env(TEST_SETTING)
    except MissingConfigError:
        result = "OK"
    print(f"case 1 (missing setting) -> {result}")

def test_empty_setting() -> None:
    os.environ[TEST_SETTING] = ""
    result = "WRONG"
    try:
        require_env(TEST_SETTING)
    except MissingConfigError:
        result = "OK"
    print(f"case 2 (empty setting)   -> {result}")

def test_config_loads() -> None:
    os.environ["OPENAI_API_KEY"] = "sk-test-123"
    config = load_config()
    result = "OK"
    if config.openai_api_key != "sk-test-123":
        result = "WRONG"
    if config.max_iterations <= 0:
        result = "WRONG"
    print(f"case 3 (config loads)    -> {result}")

if __name__ == "__main__":
    test_missing_setting()
    test_empty_setting()
    test_config_loads()
```

```
# tests/unit/test_tool_bridge.py
# Checks the schema bridge with tiny fakes — no service, no network.
# Run from the repo root: python -m tests.unit.test_tool_bridge
from src.tool_bridge import extract_text, mcp_tools_to_openai_schema

class FakeTool:
    def __init__(self, name, description, input_schema):
        self.name = name
        self.description = description
        self.input_schema = input_schema

class FakeBlock:
    def __init__(self, text):
        self.text = text

class FakeResult:
    def __init__(self, texts):
        self.content = []
        for text in texts:
            self.content.append(FakeBlock(text))

def test_schema_shape() -> None:
    tool = FakeTool("get_service_status", "Get status.",
                    {"type": "object", "properties": {}})
    schemas = mcp_tools_to_openai_schema([("status__get_service_status", tool)])
    one = schemas[0]
    result = "OK"
    if one["type"] != "function":
        result = "WRONG"
    if one["function"]["name"] != "status__get_service_status":
        result = "WRONG"
    if "properties" not in one["function"]["parameters"]:
        result = "WRONG"
    print(f"case 1 (openai schema shape)   -> {result}")

def test_missing_description() -> None:
    tool = FakeTool("t", None, {"type": "object"})
    schemas = mcp_tools_to_openai_schema([("s__t", tool)])
    result = "OK"
    if schemas[0]["function"]["description"] != "":
        result = "WRONG"
    print(f"case 2 (None description -> '') -> {result}")

def test_extract_text_joins() -> None:
    result_obj = FakeResult(["line one", "line two"])
    text = extract_text(result_obj)
    result = "OK"
```

```
    if text != "line one\nline two":
        result = "WRONG"
    print(f"case 3 (extract_text joins)    -> {result}")

if __name__ == "__main__":
    test_schema_shape()
    test_missing_description()
    test_extract_text_joins()
```

```
# tests/integration/test_agent.py
# Launches the real example services over MCP. Each case prints OK or WRONG.
# Run from the repo root: python -m tests.integration.test_agent
import asyncio

from src.agent import run_agent
from src.config import load_config
from src.models.llm import create_client
from src.server_pool import MCPServerPool

STATUS = {"name": "status", "command": "python",
          "args": ["servers/status_server.py"]}
RUNBOOKS = {"name": "runbooks", "command": "python",
            "args": ["servers/runbook_server.py"]}
BROKEN = {"name": "status", "command": "python",
          "args": ["servers/does_not_exist.py"]}

async def connect(servers):
    config = load_config()
    client = create_client(config)
    pool = MCPServerPool(servers, connect_timeout=config.connect_timeout)
    await pool.connect_all()
    return config, client, pool

async def test_discovery_namespaced() -> None:
    config, client, pool = await connect([STATUS, RUNBOOKS])
    names = []
    for namespaced, tool in pool.named_tools:
        names.append(namespaced)
    result = "OK"
    if "status__get_service_status" not in names:
        result = "WRONG"
    if "runbooks__search_runbooks" not in names:
        result = "WRONG"
    await pool.aclose()
    print(f"case 1 (tools discovered + namespaced) -> {result}")
```

```
async def test_routing_across_servers() -> None:
    config, client, pool = await connect([STATUS, RUNBOOKS])
    task = ("Is checkout down right now, and what does the runbook say to "
            "do about it?")
    out = await run_agent(task, pool, client, config.max_iterations)
    tools_used = []
    for step in out.steps:
        tools_used.append(step["tool"])
    result = "OK"
    if "status__get_service_status" not in tools_used:
        result = "WRONG"
    if "runbooks__read_runbook" not in tools_used:
        result = "WRONG"
    await pool.aclose()
    print(f"case 2 (routes to both services)       -> {result}")

async def test_fallback_when_status_down() -> None:
    config, client, pool = await connect([BROKEN, RUNBOOKS])
    result = "OK"
    if "status" in pool.sessions:
        result = "WRONG"
    if "runbooks" not in pool.sessions:
        result = "WRONG"
    out = await run_agent("Is checkout down, and what does the runbook say?",
                          pool, client, config.max_iterations)
    if not out.answer:
        result = "WRONG"
    await pool.aclose()
    print(f"case 3 (status down, runbooks only)    -> {result}")

async def main() -> None:
    await test_discovery_namespaced()
    await test_routing_across_servers()
    await test_fallback_when_status_down()

if __name__ == "__main__":
    asyncio.run(main())
```

Expected unit output (all `OK`):

```
case 1 (missing setting) -> OK
case 2 (empty setting)   -> OK
case 3 (config loads)    -> OK
```

```
case 1 (openai schema shape)   -> OK
case 2 (None description -> '') -> OK
case 3 (extract_text joins)    -> OK
```

Expected integration output (launches the real services over MCP):

```
case 1 (tools discovered + namespaced) -> OK
case 2 (routes to both services)       -> OK
case 3 (status down, runbooks only)    -> OK
```

That's the whole project. If your version discovers every tool from the services (none hardcoded), routes the incident question to both services, and still answers when the status service is down — you've built it.